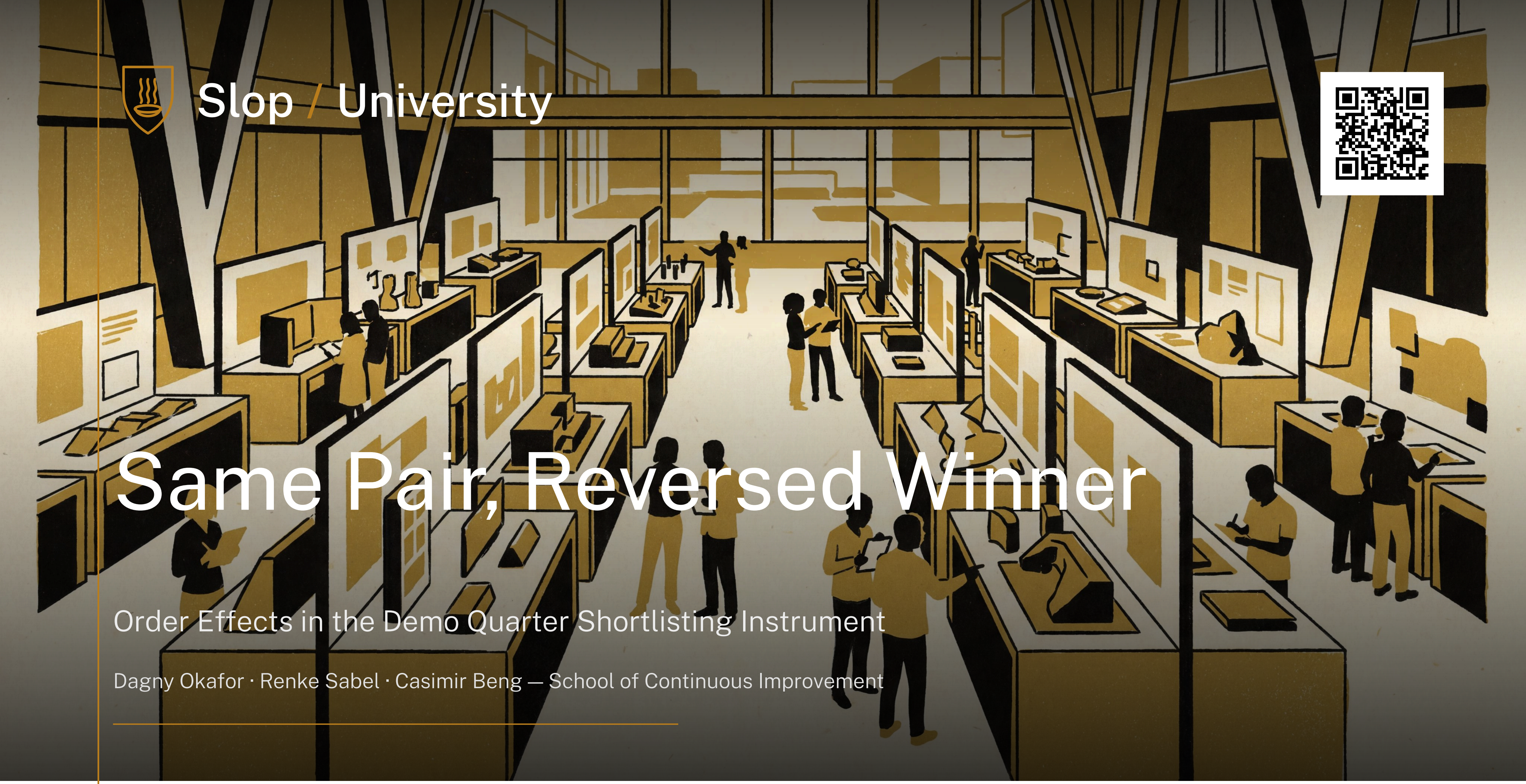

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Same Pair, Reversed Winner

Order Effects in the Demo Quarter Shortlisting Instrument

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Motivation

Demo Quarter has shortlisted showcase entries with a pairwise-comparison instrument since 2025, echoing the pairwise-judgment tools now standard in AI-system evaluation. Whether its rankings survive a basic manipulation — reversing which entry appears first — has not been tested.

Research questions

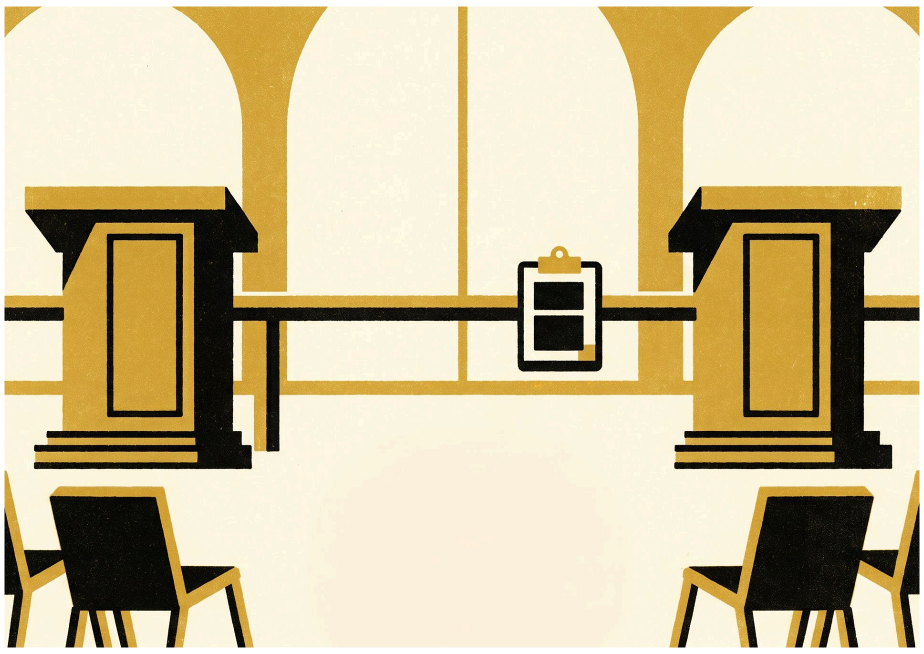
- Does reversing presentation order flip the recorded winner?
- How much of the live shortlist would change under reversed order?
- Where on the margin scale does order sensitivity concentrate?

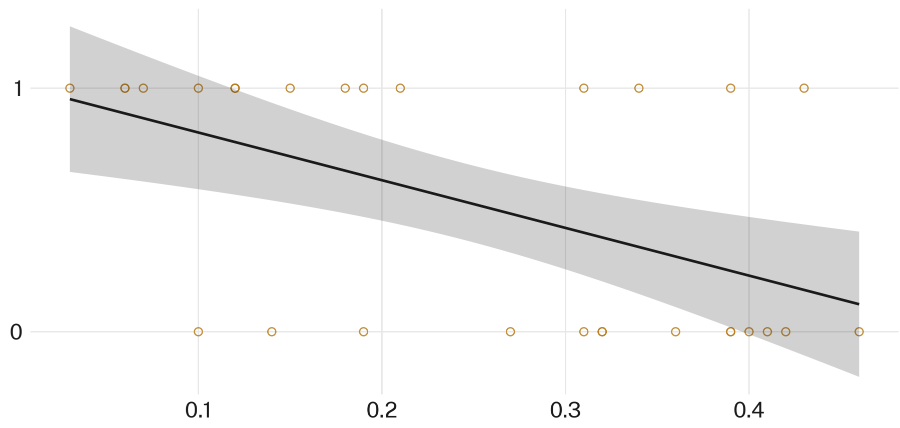
Study design

- n = 286 live pairwise judgments · 6 Demo Quarter cycles · Mar 2025–Feb 2026
- Bradley–Terry ranking model; every pair re-scored in reversed presentation order by the same rater pool
- Rater identity and instrument thresholds held constant; only display order varied

What we found

Reversing order flipped the recorded winner in 34% of re-scored pairs (97/286). Pairs decided by a margin under 0.15 flipped nearly ten times as often as those decided above 0.3.





Flip rate fell from 58% of pairs decided under a 0.15 win-probability margin to 6% of pairs decided above 0.3 (n = 286, six cycles, Mar 2025–Feb 2026); the fitted trend crosses 50% at a margin of 0.09.

Implications

Projected onto the live shortlist, 5 of 24 Demo Quarter slots (21%) would have gone to a different entry under reversed order. The instrument is least stable exactly where it carries the most weight — close calls near the shortlist cutoff, where a slot is decided by whichever entry the panel happened to see first. Wide-margin pairs, by contrast, held their winner almost without exception, suggesting the instrument is trustworthy exactly where it is least needed. Next: piloting a fixed double-pass protocol — scoring every pair in both orders and averaging the result — ahead of the coming cycle.

Selected references

1. Bradley, R. A., & Terry, M. E. (1952). Rank analysis of incomplete block designs: I. The method of paired comparisons. *Biometrika*, 39(3/4), 324–345. doi:10.2307/2334029
2. Thurstone, L. L. (1927). A law of comparative judgment. *Psychological Review*, 34(4), 273–286. doi:10.1037/h0070288
3. Pollitt, A. (2012). The method of Adaptive Comparative Judgement. *Assessment in Education: Principles, Policy & Practice*, 19(3), 281–300. doi:10.1080/0969594X.2012.665354
4. Wang, P., et al. (2023). Large language models are not fair evaluators. arXiv:2305.17926.
5. Zheng, L., et al. (2023). Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. arXiv:2306.05685.

Rater identities de-identified; no showcase entrant score adjusted retrospectively.